Supplementary material for: Reference standards built by editing automatic pre-scoring were closer to the assisting algorithm than the same experts' unassisted scoring: a paired study of polysomnography scoring

Quoc Viet Nguyen, Duc Hoai Anh Nguyen, Phuc Tai Duong,
Dzung Van Dinh and Zengchang Qin

This document contains the Supplementary Methods, Supplementary Tables S1–S18 and Supplementary Figure S1. The analysis plans are provided separately as Supplementary File S1.

Supplementary Methods

The pre-fill algorithms

The pre-fills were produced by research algorithms developed by the dataset's first author and colleagues: for sleep staging, a deep-learning stager [1]; for EEG arousals, a detector whose reviewed output had earlier produced part of the center's clinical arousal references [2]; for limb movements, a detector described in [3, 4]; and for respiratory events, a knowledge-based detector [5]. The dataset documentation gives no version numbers.

Agreement and consensus details

Event tasks used only their own labels. For respiratory events, obstructive, central and mixed apneas, hypopneas and central hypopneas were pooled, and respiratory-effort-related arousals were excluded. Events were matched one to one, overlapping pairs with the largest overlap first; two empty scorings counted as F1 = 1, and one empty scoring against a non-empty one as F1 = 0. Consensus events were formed from contiguous 0.1-second bins without a minimum-duration filter. A robustness check (amendment 005, C6) dropped consensus events, full and leave-one-out, shorter than the scoring rules' minimum durations: 3 s for arousals, 10 s for respiratory events and 0.5 s for limb movements; the scorings and the pre-fill were left unchanged.

Secondary analyses: operational details

Crossed-bootstrap intervals used 10,000 resamples for the primary and 2,000 for the secondary analyses, as planned. Pairwise percent agreement was computed on epochs for stages and on 0.1-second bins for events. Events per hour of sleep used the dataset's contextual sleep stages. The acceptance lift by manual agreement used four groups of the number of the 12 manual scorings that shared the most frequent label (12, 10–11, 8–9, and 7 or fewer). The four robustness variants were a plurality stage consensus, a 9-of-12 supermajority, event matching that requires an intersection over union of at least 0.2, and percent agreement for stages. Experts who anchored more in one task tended to anchor more in the others (Spearman +0.29 to +0.55 across the six task pairs); this assumes that scorer indices denote the same person in every task, which the documentation does not confirm.

Labels that most experts rejected. The first amendment's rule (R4) selected units where at least 10 of the 12 manual scorings, the scored expert's included, rejected the pre-fill's label.

Because the scored expert's own manual label enters the selection, a positive lift is partly expected from regression to the mean even without anchoring. The fourth amendment's rule (C1) selects, for each expert, the units where at least 9 of the other 11 experts' manual labels reject the pre-fill's label. For stages, another expert's label counts as rejecting only if it is present and differs from the pre-fill's, and an expert contributes an epoch only if they labelled it in both sessions. For events, the units are the pre-fill's events after clipping to the window; a manual scoring rejects an event if none of its events overlaps it, and an expert adopts it if their assisted scoring overlaps it. The exact sign test for C1 counts the units with a nonzero mean lift; the one unit whose mean lift is exactly zero (a limb-movement recording) is a tie and is dropped (retaining it as a non-success would give a larger p). If each expert's two sessions were exchangeable and experts labelled independently given the unit within a session, the lift would be zero in expectation under C1. Table S3 gives both rules.

Independent stagers and the negative control

YASA 0.7.0 [6] used its default model with the C4-M1 EEG channel (Cz-M1 for the one recording without it), E1-M2 and chin electromyography. GSSC 0.0.9 [7] ran in its own environment on the CPU (PyTorch 2.5.1; later versions refuse to load its saved networks) with its default networks and settings, which filter to 0.3–30 Hz, resample to 2,560 samples per 30-second epoch and drop a trailing partial epoch; it used the same EEG and E1-M2 channels. Both stagers' labels start at the beginning of the recording and were mapped to epochs by the window rule of the primary analysis. The settings were fixed before the stagers were run on PSG-IPA; GSSC was first checked on synthetic signals.

For the negative control (C2), the units were epochs where a stager and the pre-fill both give a label and the labels differ. Movement toward a target was each expert's share of assisted labels equal to the target minus the share of manual labels equal to it, on those epochs; side-taking toward a stager compared, among those epochs where both consensuses exist and differ, how often each consensus equals the stager. Common support (A1, C3) is the set of epochs where both 12-expert consensuses exist; the session-free contrast is Δ_{auto} minus the stager's change on epochs where the pre-fill and the stager are both labelled and both consensuses exist (Table S11).

Audit designs

The population of the audit emulations was the 4,857 staging epochs on which the pre-fill and YASA are labelled and the manual consensus exists. Audits of blocks (C4) divided each recording's analysis window into consecutive 30-minute blocks of 60 epochs from lights off (87 blocks holding population epochs) and drew a simple random sample of blocks within each recording, $\max(1, \text{round}(f \times \text{blocks}))$ of them. The audit alone (D1) expands the audited block totals to the population mean with the known population size; the difference estimator (D2) adds the expanded audited totals of unassisted minus assisted agreement to the known population mean of assisted agreement. Both are unbiased, and their variances are exact for stratified simple random sampling of blocks. Audits of single epochs (B1) drew epochs instead of blocks with proportional allocation [8]; they treat neighboring epochs as independent units, which experts, who score an epoch in the context of its neighbors, cannot do in practice. The model's ranking probabilities (B2, Table S8) also treat epochs as independent and are optimistic for serially correlated sleep data. Treating the epochs with an unassisted consensus as known slightly flatters the difference estimator.

The simulation

Model. We modeled sleep staging with a two-label latent-class model [9]. Each epoch has a true label and one competing label. A share w of epochs is clear, and all raters label them correctly; every other epoch has a difficulty $b = -|x|$, with $x \sim N(\mu, \sigma^2)$, and a rater with ability θ gives

the true label with probability logistic($\theta - b$). Experts have ability 0, the pre-fill δ_A, and an independent competitor δ_B. Raters are independent given the epoch. In the assisted session each expert forms a judgment of the epoch, a new draw with the same accuracy or, in a sensitivity analysis, a repeat of the manual label with probability c; if the judgment matches the pre-fill the expert keeps it, and otherwise switches to the pre-fill with probability α. All quantities are exact large-sample expectations of percent agreement, computed by numerical integration over difficulty. Because the difficulty is folded, experts are right more often than wrong on every epoch, so the model's truth is what a very large unassisted panel would converge to, the standard the audit also uses.

Calibration. The parameters (w, μ, σ) were fitted by least squares to the shares of epochs at each manual agreement level and to the mean pairwise manual agreement [10]; δ_A by bisection on the experts' agreement with the pre-fill; and α by an identity of the model, under which anchoring equals α times one minus the experts' agreement with the pre-fill. The model was calibrated to the 5,103 stage epochs labelled by all scorings. It overstated the mean pairwise manual agreement slightly (0.830 against 0.821), because some epochs had three or four distinct manual labels, which two labels cannot represent. As an internal check, the study's own estimators were applied to 400,000 epochs simulated at the calibrated point and had to agree with the analytic values within 0.015; they agreed within 0.001.

Consequences. For references of 1 to 25 experts, under majority and three-quarter supermajority rules, we computed the reference's accuracy against the truth; the pre-fill's measured and true accuracy, whose difference on the epochs the reference includes is its bias; the handicap, the bias in favor of the pre-fill over a competitor that is equally accurate and has the experts' error pattern; and the break-even advantage, how much more accurate than the pre-fill such a competitor must be to tie it. A grid crossed α from 0 to 0.8 with δ_A from -1 to 3; the break-even advantage was undefined (no competitor could tie) in 508 of the 720 cells with $\alpha > 0$. Sensitivity analyses varied expert ability, intra-rater consistency, idiosyncratic pre-fill errors and expert-specific anchoring.

Results. In the calibrated model, 8.5% of epochs were clear, the pre-fill was about as accurate as an average expert (true accuracy 0.882 against 0.880), and the anchoring strength was $\alpha = 0.31$. The model matched the ambiguity gradient and the human change, but overpredicted the pre-fill's gain (+0.065 against +0.045) and the adoption of strongly rejected labels (+0.332 against +0.228; Table S5): the real pre-fill errs less often than an expert-level model rater on epochs that experts find clear, and experts may resist it more when they disagree strongly. The model shows three things (Figure S1; Table S6). First, adding experts moved a manual reference toward the truth (the pre-fill's bias went from -0.051 with one expert to -0.004 with 25) but moved an assisted reference away from it (+0.001, +0.050 and +0.062 with 1, 12 and 25 experts), and 12 assisted experts formed a less accurate reference than 12 manual experts (0.944 against 0.970). Second, a 9-of-12 consensus cut the bias on the epochs it kept to +0.017 but excluded 14% of epochs. Third, anchoring and the pre-fill's gain were positive at every pre-fill ability in the grid, so they cannot show whether an assisted reference became more or less accurate; with 12 experts, the assisted reference became the more accurate of the two from a pre-fill offset of +1.0 (true accuracy 0.945), while the human change turned positive only from +1.5 (true accuracy 0.965); the observed human change for stages (-0.016 over all epochs, -0.010 on common support; kappa, where the model's crossovers are in percent agreement, the sign being the same in either) lies on the negative side of both crossovers. Every sensitivity analysis kept these conclusions (Table S7).

The experts' own consensus. Judged against the assisted consensus, the experts' own manual consensus was closer than the pre-fill (percent agreement 0.964 against 0.947 for stages; the pre-fill was closer in only 6 of 20 units), which contradicted our prediction (A3). The calibrated anchoring strength could not reproduce side-taking and this result together. A configuration examined after the independent-rater checks, and not pre-specified, with weaker anchoring ($\alpha = 0.22$) and intra-rater consistency $c = 0.5$, reproduced the stage side-taking share (0.926 against 0.930 observed), the experts' agreement across sessions (0.964 against 0.964) and the

pre-fill's gain (+0.046 against +0.045). Its anchoring is 0.037 against the observed 0.052, which the calibrated configuration fits by construction; its bias with 1, 12 and 25 assisted experts is −0.014, +0.032 and +0.041 (calibrated: +0.001, +0.050 and +0.062), so the bias grows with the panel in both configurations; a competitor needed a true accuracy +0.076 above the pre-fill's to tie it, whereas at the calibrated strength no competitor could tie; and its validation targets are in Table S5. The two configurations each miss one target, the calibrated one the experts' own consensus (A3) and the exploratory one the anchoring measure, so they are illustrations under the model, not an uncertainty interval (amendment 006).

Run history. The first run stopped at calibration: manual agreement statistics cannot distinguish an epoch on which experts are mostly right from one on which they are mostly wrong, and the unconstrained fit exploited this symmetry. A dated note introduced the folded difficulty before the run was repeated.

Review of FDA clearances

Two authors checked the AI-assisted coding of the 25 summaries against the source documents (P.T.D. 12 summaries, D.H.A.N. 13), recording for every coded field whether they agreed, and whether the summary said that the reference scorers had worked from automatic pre-scoring by any software. Of the 100 coded fields, they agreed with 89, partly agreed with 5 and disagreed with 6. Their check corrected the coding of four summaries, and the AI assistant reread each cited passage and confirmed the correction. The 2024 Dreem 3S clearance (K242094) reports no new performance testing, because it only adds a predetermined change control plan to K223539, so it left the clearances with performance data. The reference of the first SOMNUM clearance (K223922), first coded as not described, is a 2-of-3 majority of three technologists whose independence is not stated. The Nox Sleep System study (K192469) exported the routine clinical scoring and then ran the device on the same recordings, so blinding to the device is implied by the design. The 2012 Sleep Profiler summary (K120450) names three raters. The check also made consensus rules and notes more precise without changing any count. No summary said whether the reference scorers worked from automatic pre-scoring by any software. The 2026 SOMNUM summary (K253390) stated that its scorers had no access to the device's output, which excludes the device's own pre-scoring but not that of other software; the Hypnos summary (K250851) used another algorithm's unedited output, so the question does not arise. A keyword search of all 25 summaries for wording on pre-scoring, editing, assisted scoring, blinding and access found no other relevant statement; several summaries describe manual review of automatic scoring as the device's intended use, which concerns the device, not the reference.

Blind second coding (amendment 005, D1). Two co-authors (P.T.D. and D.H.A.N.) then re-coded the three key fields of the 22 summaries with performance data (reference type, blinding to the device's output and pre-scoring) from the definitions of a coding sheet, by hand and, by their own account, without consulting any code (D.H.A.N. held the team's manuscript package, whose Table S9 lists the first-round reference and blinding codes), and sent their sheets to the corresponding author. Each had checked part of the 22 in the first round (P.T.D. 10 of the 12 summaries they had checked, D.H.A.N. 12 of 13) and had then seen the AI's codes for them, which the first-round records show and both confirmed when asked, so each was blind only for the other's summaries. Every quoted passage was located in its summary (four of one coder's quotations were not verbatim for recorded reasons, and one of the other's was corrected to the summary's wording after the authors queried it). Table S17 gives the agreement with the AI's checked coding (the first coding), overall and by exposure: P.T.D. agreed on 21/22, 22/22 and 22/22 summaries for the three fields and D.H.A.N. on 18/22, 20/22 and 20/22 (kappa 0.636 to 1.000); on the summaries for which they were blind, P.T.D. agreed on 11/12, 12/12 and 12/12 and D.H.A.N. on 8/10, 8/10 and 8/10. The two coders agreed with each other on 19/22, 20/22 and 20/22 (the first coder's sheet reached the team channel before the second had finished; the second states that they did not open it). The 8 fields on which a coder differed from the first coding, in

5 summaries, are listed in Table S18 with their adjudication by the corresponding author, who had directed the first coding but had not coded, from the summary text on 28 October 2026, following a written guide that set out each item's passages and the category definitions (the guide, the passages and both coders' quotations are in the study repository, so the adjudication can be redone): 1 code changed, the SomnoMetry reference (K221179), from routine clinical scoring to not described, the code both coders had chosen, because no sentence of that summary says who scored the archived clinical recordings, when or how. The counts in Table S9 and in the Results follow the adjudicated coding (routine clinical scoring 4 to 3 summaries, not described 2 to 3). The statement that no summary reported pre-scoring stands; the coded passage of K221179 now carries its "AASM gold standard of manually scored PSG data", so 10 coded passages call the reference manually scored. The coders and the adjudicator were authors, not blind to the study's hypothesis; each coder was blind to the first-round codes for the other's summaries only.

Software and verification

The analyses used Python 3 with the standard library; YASA ran with MNE-Python [11] and GSSC with PyTorch, each in a separate environment. Every result is generated from the aggregate reports of the analysis code, which also hold pre-specified quantities that the tables omit for space (the audit emulations at 1%, 2% and 20%, the number of expert-unit pairs behind each C1 value, the per-unit C1 lifts and C6 differences, the per-recording values of the experts'-own-consensus check (A3), and the percent-agreement values and the difference-in-differences of the common-support check). The study's Cohen's kappa matched scikit-learn 1.9.1 over all 130 staging comparisons with the pre-fill (largest difference 2.2×10^{-16}), and its exact sign tests matched SciPy 1.17.1's binomial test for all 5 sets of units (largest difference 0). For amendment 005 D2, a co-author (D.H.A.N.) who had not read the analysis code re-implemented M1 and M2 for sleep staging from the text of protocol 001 and the PSG-IPA annotations, using the OpenAI Codex coding assistant, with five synthetic tests including 50 random kappa pairs against scikit-learn; the implementer held the 19 October manuscript package with its result tables and per-recording figure, and the assistant had seen some of those tables; the implementer disclosed both, and neither had the analysis code. All 20 per-recording values (mean ΔA, Δ_{auto}, Δ_{human} and DiD for SN1 to SN5) agreed with the study's within 2.3×10^{-16}, inside the amendment's 0.001 rule. At the request of the pre-submission review of 26 October 2026, the same co-author then re-implemented M1 for the three event tasks from the protocol's matching rule (one-to-one matching of overlapping events, largest overlaps first), with six synthetic tests per task and one end-to-end test; all 15 per-recording values and the 36 per-expert means (Table S16) agreed with the study's within 5.6×10^{-17}. The protocol does not fix the order among candidate pairs with equal overlap; both implementations take the earlier-starting events first, so this agreement does not test that convention. The event-task consensus and inflation measures, side-taking, M3 to M5, the audit emulation, the simulation and the stager comparisons were not re-implemented and rest on the analysis code's synthetic tests. For the second half of D2, the same co-author re-ran the whole pipeline on 29 October 2026 from a fresh clone of the study repository on a freshly downloaded copy of PSG-IPA (all 1,025 files verified against the published checksums), on Windows with Python 3.11.9: first with the standard-library analyses alone and then with the two stager environments rebuilt from the pinned requirements (two dependencies of the YASA environment resolved above their pins, pandas 3.1.0 for 3.0.6 and antropy 0.2.3 for 0.2.2; both label files were nonetheless reproduced epoch for epoch). The generated values file as it then stood was reproduced identically, the YASA and GSSC per-epoch labels were reproduced identically, and the aggregate reports agreed with the committed ones to within 2.2×10^{-16}, the last floating-point bit, consistent with Python's per-process hash randomisation, which changes the order of the expected-agreement sum in the kappa computation (verified for the protocol 001 report by running it under two hash seeds). The comparison tool and its output are in the repository; the environment listings are held by the authors.

Supplementary Tables

Table S1: Names used in the text and the codes used in the analysis plans (Supplementary File S1).

Plan code	Name in the text
M1	Anchoring (ΔA)
M2	Reference inflation (DiD = $\Delta_{\text{auto}} - \Delta_{\text{human}}$)
M3	Agreement among scorers where the pre-fill agrees or disagrees with the manual consensus
M4	Clinical indices
M5	Independent stager (YASA)
R1, R2	Acceptance lift by pre-fill stage; anchoring by respiratory event type
R3, R4	Acceptance lift by manual agreement; adoption of labels at least 10 of 12 experts rejected
R5, R6	Consistency of experts across tasks; robustness variants
A1, A2, A3	Common support; side-taking; the experts' own consensus
B1, B2	Audit of single epochs on PSG-IPA; audit in the calibrated model
D0, D0s, D1, D2	Assisted reference; 9-of-12 assisted reference; audit alone; difference estimator
C1–C4, V1–V10	Calibration targets; validation targets of the simulation (amendment 003)
C1 (amendment 005)	Adoption of labels that at least 9 of the other 11 experts rejected
C2, C3	Negative control toward the stagers; the pre-fill's gain on common support
C4, C5, C6	Audit of blocks; second stager (GSSC); consensus events of the rules' minimum duration
D1, D2, D3 (amendment 005)	Blind second FDA coding; independent re-implementation; library cross-check

Table S2: Acceptance lift by the pre-fill's sleep stage (R1) and anchoring by respiratory event type (R2).

Stratum	Lift or anchoring (95% interval)
W	+0.029 (+0.001 to +0.078)
N1	+0.173 (+0.098 to +0.254)
N2	+0.040 (−0.021 to +0.111)
N3	+0.095 (+0.051 to +0.148)
R	+0.054 (+0.016 to +0.098)
Respiratory, apneas	+0.056 (−0.108 to +0.202)
Respiratory, hypopneas	+0.234 (+0.156 to +0.309)

R1: P(assisted label = pre-fill) minus P(manual label = pre-fill), for epochs the pre-fill labels with that stage.
R2: M1 anchoring in event F1.

Table S3: Dependence on ambiguity (R3) and adoption of pre-fill labels that most experts rejected manually: selected with the scored expert's own label (R4) and without it (C1).

Task	R3 12 (unanimous)	R3 10–11	R3 8–9	R3 ≤7	R4 ≥10 of 12	C1 ≥9 of other 11
Sleep stages	−0.005 (−0.016 to +0.002)	+0.068 (+0.024 to +0.119)	+0.149 (+0.081 to +0.225)	+0.191 (+0.104 to +0.292)	+0.241 (+0.139 to +0.371)	+0.169 (+0.074 to +0.279)
EEG arousals	−0.002 (−0.003 to −0.001)	+0.062 (+0.026 to +0.094)	+0.124 (+0.064 to +0.173)	+0.182 (+0.106 to +0.243)	+0.281 (+0.188 to +0.380)	+0.269 (+0.160 to +0.390)
Respiratory events	+0.003 (0.000 to +0.006)	+0.111 (+0.071 to +0.150)	+0.216 (+0.145 to +0.288)	+0.240 (+0.129 to +0.359)	+0.283 (+0.186 to +0.385)	+0.231 (+0.120 to +0.351)
Limb movements	0.000 (0.000 to 0.000)	+0.126 (+0.087 to +0.164)	+0.273 (+0.214 to +0.327)	+0.345 (+0.278 to +0.409)	+0.514 (+0.404 to +0.611)	+0.374 (+0.103 to +0.647)

Values: P(assisted = pre-fill) minus P(manual = pre-fill), mean over experts, with 95% crossed-bootstrap intervals. R3 bins count the manual experts sharing the most frequent label of a unit; R3 and R4 use epochs for stages and 0.1-second bins for events. R4 selects units with the scored expert's own manual label, so part of its lift is expected from regression to the mean. C1 selects units with the other 11 experts' labels only, and counts whole pre-fill events for the event tasks; R4's rule applied to whole events gives +0.303 (arousals), +0.258 (respiratory events) and +0.499 (limb movements).

Table S4: Robustness of the primary conclusions to analysis choices (R6, C6).

Variant	Anchoring units > 0	Inflation units > 0	Task DiD range
Plurality stage consensus	19/20 ($p = 4.0 \times 10^{-5}$)	20/20 ($p = 1.9 \times 10^{-6}$)	0.054 to 0.178
Supermajority, 9 of 12 (9 of 11)	19/20 ($p = 4.0 \times 10^{-5}$)	19/20 ($p = 4.0 \times 10^{-5}$)	0.046 to 0.114
Event matching at IoU ≥ 0.2	19/20 ($p = 4.0 \times 10^{-5}$)	19/20 ($p = 4.0 \times 10^{-5}$)	0.059 to 0.178
Percent agreement for stages	19/20 ($p = 4.0 \times 10^{-5}$)	20/20 ($p = 1.9 \times 10^{-6}$)	0.054 to 0.178
Consensus events of the rules' minimum duration (C6)	unchanged	15/15 event units ($p = 6.1 \times 10^{-5}$)	+0.054 to +0.207 (event tasks)

C6 drops consensus events shorter than 3 s (arousals), 10 s (respiratory events) or 0.5 s (limb movements); anchoring does not involve a consensus.

Table S5: Validation of the two model configurations against observations they were not fitted to (sleep stages, percent agreement, pooled over epochs).

Target	Model (calibrated)	Model (exploratory)	Observed
Δ_{auto} (V1)	+0.065	+0.046	+0.045
Δ_{human} (V2)	−0.011	−0.005	−0.012
DiD (V3)	+0.076	+0.051	+0.057
M3, pre-fill agrees: manual (V4)	0.869	0.869	0.872
M3, pre-fill agrees: assisted (V4)	0.897	0.890	0.915
M3, pre-fill disagrees: manual (V4)	0.636	0.636	0.590
M3, pre-fill disagrees: assisted (V4)	0.529	0.528	0.526
R3 lift, 12/12 agree (V5)	−0.006	−0.002	−0.007
R3 lift, 10–11/12 (V5)	+0.069	+0.049	+0.065
R3 lift, 8–9/12 (V5)	+0.145	+0.101	+0.146
R3 lift, ≤7/12 (V5)	+0.177	+0.123	+0.179
R4 lift (V6)	+0.332	+0.230	+0.228
Share of R4 units (V7)	0.043	0.043	0.028
Pre-fill vs manual consensus (V8)	0.886	0.886	0.898
Change in YASA's agreement, manual to assisted consensus (V9)	−0.013	−0.006	negative (−0.040 kappa)
Growth of the pre-fill's lead over YASA (V10)	+0.078	+0.052	positive (+0.115 kappa)

The only anchoring parameter, α, was set from the pooled anchoring measure (C4). V9 and V10 are compared by sign only: their observed values are kappa (YASA), the model's percent agreement. Exploratory: the configuration examined after the independent-rater checks ($\alpha = 0.22$, consistency 0.5; amendment 006), whose anchoring M1 is 0.037 against the observed 0.052; the calibrated configuration fits the pooled anchoring measure by construction.

Table S6: Consequences in the calibrated model: references built from manual versus assisted scorings.

Experts, rule	Reference accuracy manual	Reference accuracy assisted	Pre-fill measured (bias) manual	Pre-fill measured (bias) assisted	Handicap manual	Handicap assisted
1, majority	0.880	0.881	0.831 (−0.051)	0.883 (+0.001)	−0.001	+0.050
3, majority	0.920	0.912	0.857 (−0.026)	0.915 (+0.033)	−0.001	+0.063
5, majority	0.936	0.922	0.865 (−0.017)	0.926 (+0.044)	−0.001	+0.068
12, majority	0.970	0.944	0.886 (−0.005)	0.951 (+0.050)	0.000	+0.064
12, 9 of 12	0.996	0.983	0.926 (−0.001)	0.985 (+0.017)	0.000	+0.020
25, majority	0.970	0.935	0.878 (−0.004)	0.944 (+0.062)	0.000	+0.078

True accuracy of the pre-fill 0.882; of the average expert 0.880. Bias: measured minus true accuracy on the epochs the reference includes. Handicap: the bias in favor of the pre-fill over an equally accurate competitor with the experts' error pattern.

Table S7: Sensitivity analyses of the simulation: 12 assisted experts, majority consensus.

Model	Pre-fill bias	Handicap	Break-even advantage
Base	+0.050	+0.064	none
S1: expert ability spread 0.5	+0.052	+0.067	none
S2: intra-rater consistency 0.5	+0.050	+0.064	none
S3: idiosyncratic pre-fill errors	+0.050	+0.064	none
S4: expert-specific anchoring	+0.044	+0.056	+0.111

Break-even: the true-accuracy advantage a competitor needs to tie the pre-fill; "none" means even a perfect competitor falls short.

Table S8: Probability that a truly better competitor is ranked above the pre-fill (calibrated model).

Advantage	Epochs	D0	D0s	D1 5%	D2 5%	D1 10%	D2 10%
0.00	5,000	0.000	0.000	0.500	0.500	0.500	0.500
0.00	50,000	0.000	0.000	0.500	0.500	0.500	0.500
0.01	5,000	0.000	0.000	0.620	0.624	0.668	0.672
0.01	50,000	0.000	0.000	0.834	0.840	0.915	0.920
0.02	5,000	0.000	0.000	0.732	0.733	0.809	0.811
0.02	50,000	0.000	0.000	0.975	0.976	0.997	0.997
0.05	5,000	0.000	0.000	0.945	0.933	0.988	0.983
0.05	50,000	0.000	0.000	1.000	1.000	1.000	1.000

With advantage 0, the entries give the probability of ranking an equally accurate competitor above the pre-fill.

Table S9: How the 22 FDA clearances of automatic PSG scoring software (product code OLZ, 2012 to 2026) with clinical performance data built their reference, as stated in the public 510(k) summaries.

510(k)	Year	Device	Reference	Scorers	Scorers blinded to the device output
K120450	2012	Sleep Profiler	Consensus, independence not stated	3	not stated
K142825	2015	Zmachine	Independent experts, majority	2 to 4	not stated
K153412	2016	Sleep Profiler	Consensus, independence not stated	3	not stated
K153353	2016	Sandman Elite	Independent experts, majority	3	implied by design
K162140	2016	RemLogic	Independent experts, majority	3	implied by design
K162627	2017	EnsoSleep	Independent experts, majority	3	stated (object unspecified)
K163617	2017	REMbrandt	Independent experts, majority	3	implied by design
K173366	2018	NicoletOne	Independent experts, majority	3	implied by design
K182227	2019	Esprit Nova	Panel, daily practice	3	not stated
K192469	2019	Nox Sleep System	Routine clinical scoring	–	implied by design
K202142	2020	Sleepware G3	Not described	–	not stated
K210034	2021	EnsoSleep	Independent experts, majority	3 of 9	stated (object unspecified)
K221179	2022	SomnoMetry	Not described	–	not stated
K223922	2023	SOMNUM	Consensus, independence not stated	3	not stated
K223539	2023	Dreem 3S	Consensus, independence not stated	–	not stated
K233438	2024	SleepStageML	Independent experts, majority	3	not stated
K241960	2025	DeepRESP	Routine clinical scoring	–	not stated
K250058	2025	NEAT 001	Not described	–	not stated
K252330	2025	DeepRESP	Routine clinical scoring	–	not stated
K250851	2025	Hypnos	Another algorithm's output	–	not applicable
K253668	2026	Onera SleepMap	Independent experts, majority	3	not stated
K253390	2026	SOMNUM	Independent experts, majority	3	explicit

Coding and verbatim quotes are in the study's DEVICES.csv. "Not stated" means not stated in the public summary.

Table S10: Pre-specified predictions and outcomes.

Plan	ID	Prediction	Outcome
002	R1	Largest acceptance lift for N1	supported
002	R2	Larger anchoring for hypopneas than apneas	supported
002	R3	Lift grows as manual agreement falls	supported
002	R4	Experts adopt pre-fill labels a strong manual consensus rejected	supported
002	R5	Weakly positive consistency of experts across tasks	supported
002	R6	Primary tests significant under every variant	supported
003	P1	The calibrated model reproduces the sign of every validation target	supported
003	P2	With 12 experts, the assisted bias is positive and exceeds the manual bias	supported
003	P3	The assisted bias is larger with 25 experts than with 1, and positive	supported
003	P4	In every majority cell with $\alpha > 0$ the assisted handicap exceeds the manual one (360 cells)	supported
003	P5	Signatures keep their sign at every pre-fill ability, while the assisted reference is worse at $\delta_A = -1$ and better at $\delta_A = 3$	not supported
004	A1	Δ_{human} and Δ_{YASA} negative on common support	supported
004	A2	The assisted consensus sides with the pre-fill; sign test $p < 0.05$	supported
004	A3	Stages: the pre-fill is no more than 0.01 below the manual consensus	not supported
004	PB1	The assisted reference overstates the pre-fill's agreement and its lead over YASA	supported
004	PB2	The difference estimator's SE is below the audit-alone SE everywhere, in the data and the model	not supported
004	PB3	The assisted reference almost never ranks a 2-point-better competitor first; a 10% audit does with probability above 0.8	supported
004	PB4	The 9-of-12 reference ranks between the assisted reference and the audit	not supported
005	C1	Adoption of labels the other experts rejected is positive in every task, smaller than R4 for stages; sign test $p < 0.05$	supported
005	C2	Movement toward YASA's differing labels is not positive; toward the pre-fill's it is	supported
005	C3	Δ_{auto} positive on common support in all five recordings; the session-free contrast in at least four	supported
005	C4	Block audits are less precise than single-epoch audits at the nearest fraction, and need more than 5%	not supported
005	C5	GSSC loses agreement against the assisted consensus; the pre-fill's lead over it grows	supported
005	C6	Inflation positive in every event task and in at least 13 of 15 event units	supported

Plan 002 is the first amendment, 003 the second (simulation), 004 the third (audit and independent-rater checks) and 005 the fourth (checks requested by a pre-submission review). PB2 failed only in the model, where the audit alone and the difference estimator were nearly equal. PB4 failed because the 9-of-12 reference was as poor as the majority reference. C4 failed only at the largest block fraction (50%), whose nearest single-epoch fraction was 20%; at equal fractions, block audits were always less precise, and they needed 10% of blocks.

Table S11: Negative control and comparisons with the two stagers that no expert saw (sleep stages; C2, C3, C5).

Stager	Recording	Epochs differing	Toward stager	Toward pre-fill	Consensus equals stager assisted / manual	Δ stager	Δ stager, common support	Session-free contrast
YASA	SN1	169	−0.095	+0.138	0 / 9 of 10	+0.009	+0.010	+0.033
YASA	SN2	187	−0.147	+0.166	0 / 23 of 26	−0.042	−0.027	+0.103
YASA	SN3	357	−0.109	+0.152	0 / 28 of 30	−0.047	−0.021	+0.054
YASA	SN4	206	−0.155	+0.171	0 / 29 of 29	−0.056	−0.041	+0.087
YASA	SN5	133	−0.168	+0.184	0 / 24 of 27	−0.065	−0.055	+0.129
YASA	All	1,052	−0.135 (−0.206 to −0.076)	+0.162 (+0.101 to +0.233)	0% / 93% of 122	−0.040	−0.027	+0.081
GSSC	SN1	163	−0.104	+0.139	0 / 6 of 7	+0.013	+0.020	+0.023
GSSC	SN2	176	−0.151	+0.173	0 / 26 of 28	−0.057	−0.038	+0.114
GSSC	SN3	226	−0.115	+0.177	0 / 20 of 21	−0.018	−0.006	+0.038
GSSC	SN4	138	−0.176	+0.192	0 / 21 of 21	−0.020	−0.017	+0.063
GSSC	SN5	91	−0.234	+0.234	0 / 23 of 23	−0.063	−0.054	+0.128
GSSC	All	794	−0.156 (−0.242 to −0.086)	+0.183 (+0.113 to +0.268)	0% / 96% of 100	−0.029	−0.019	+0.073

Epochs differing: epochs where the stager and the pre-fill both give a label and the labels differ. Toward: P(assisted label = target) minus P(manual label = target) on those epochs, mean over experts (pooled rows: mean over expert-recording cells with a 95% crossed-bootstrap interval). Consensus equals stager: among those epochs where both consensuses exist and differ, the number (or share) where each consensus equals the stager. Δ stager: the stager's kappa against the assisted minus the manual consensus. Session-free contrast: Δ_{auto} minus Δ stager on epochs where both are labelled and both consensuses exist (kappa). In percent agreement, the session-free contrast is +0.051 for YASA and +0.045 for GSSC (positive in 5 and 5 of 5 recordings).

Table S12: Clinical indices (events per hour of sleep) of the pre-fill and the two consensuses (M4).

Task	Recording	Pre-fill	Manual consensus	Assisted consensus	Assisted minus manual
EEG arousals	SN1	8.0	19.6	20.2	+0.6
EEG arousals	SN2	5.7	5.3	6.4	+1.1
EEG arousals	SN3	11.2	16.2	16.3	+0.1
EEG arousals	SN4	13.3	13.0	14.5	+1.5
EEG arousals	SN5	23.0	26.7	30.4	+3.7
Respiratory events	SN1	6.6	5.7	6.0 (5–15 / 5–15)	+0.3
Respiratory events	SN2	8.7	3.5	3.7 (<5 / <5)	+0.2
Respiratory events	SN3	60.1	52.8	56.5 (≥30 / ≥30)	+3.6
Respiratory events	SN4	4.2	3.3	3.2 (<5 / <5)	−0.2
Respiratory events	SN5	15.0	7.6	11.3 (5–15 / 5–15)	+3.7
Limb movements	SN1	85.1	50.8	59.9	+9.0
Limb movements	SN2	34.9	25.9	26.3	+0.4
Limb movements	SN3	61.7	60.7	60.2	−0.4
Limb movements	SN4	149.7	134.8	135.6	+0.8
Limb movements	SN5	112.5	113.2	100.6	−12.6

Hours of sleep from the dataset's context stages. Respiratory events pool apneas and hypopneas; the brackets give the adult severity category of the manual and assisted consensus (thresholds 5, 15 and 30 events per hour), which differed in 0 of 5 recordings. The pre-fill's index lay closer to the assisted than to the manual consensus in 4 of 5 respiratory and 3 of 5 limb-movement recordings, but in 0 of 5 arousal recordings, where assisted experts added arousals beyond those in the pre-fill. Descriptive.

Table S13: Agreement among the 12 experts where the pre-fill agreed or disagreed with the manual consensus (M3).

Task	Pre-fill agrees manual	Pre-fill agrees assisted	Pre-fill disagrees manual	Pre-fill disagrees assisted
Sleep stages	0.868	0.916	0.593	0.528
EEG arousals	0.977	0.983	0.647	0.604
Respiratory events	0.977	0.988	0.769	0.676
Limb movements	0.987	0.997	0.590	0.728

Pooled pairwise percent agreement among the 12 scorings of each session, on epochs (stages) or 0.1-second bins (events) where the manual consensus exists, mean over recordings.

Table S14: Audits of 30-minute blocks at five sizes, and audits of single epochs (sleep stages, C4 and B1).

Blocks audited	Share of epochs	Blocks	Gap D1 / D2	Pre-fill D2	YASA D2	Single epochs gap D2	Degrees of freedom	Gap D2, t multiplier
5%	6%	5 of 87	±0.116 / ±0.062	±0.035	±0.041	±0.039	–	–
10%	12%	10 of 87	±0.080 / ±0.045	±0.024	±0.028	±0.027	5	±0.059
20%	21%	18 of 87	±0.056 / ±0.030	±0.016	±0.019	±0.018	13	±0.033
30%	29%	25 of 87	±0.045 / ±0.024	±0.013	±0.015	–	20	±0.026
50%	51%	44 of 87	±0.028 / ±0.015	±0.008	±0.010	–	39	±0.016

95% half-widths. The bias of the assisted reference for the gap is +0.059; D2's half-width falls below it from 10% of blocks. Single epochs: amendment 004's emulation at the same fraction of epochs. The last two columns (amendment 006) multiply the gap's D2 standard error by $t_{0.975}$ with the audit's degrees of freedom (audited blocks minus the five recording strata), as an audit estimating the within-stratum variance from its own blocks would; the standard error itself still uses the population variances. Under the t multiplier the half-width falls below the bias from 10% of blocks (at 10%, 0.0586 against 0.0593).

Table S15: The 20 recording-task units: anchoring, the two consensus changes, the difference-in-differences and side-taking (M1, M2, A2).

Task	Recording	ΔA (M1)	Δ_{auto}	Δ_{human}	DiD (M2)	Side-taking difference (n)
Sleep stages	SN1	+0.095	+0.062	−0.012	+0.074	+1.000 (25)
Sleep stages	SN2	+0.112	+0.102	−0.014	+0.117	+0.946 (37)
Sleep stages	SN3	+0.065	+0.056	−0.022	+0.078	+0.733 (45)
Sleep stages	SN4	+0.055	+0.064	−0.011	+0.074	+0.882 (34)
Sleep stages	SN5	+0.078	+0.090	−0.020	+0.110	+0.867 (30)
EEG arousals	SN1	−0.002	+0.018	+0.004	+0.014	+0.158 (3,455)
EEG arousals	SN2	+0.162	+0.155	−0.035	+0.190	+0.517 (1,816)
EEG arousals	SN3	+0.073	+0.052	−0.002	+0.054	+0.650 (4,720)
EEG arousals	SN4	+0.022	+0.056	−0.043	+0.099	+0.510 (4,963)
EEG arousals	SN5	+0.075	+0.078	−0.009	+0.087	+0.760 (8,222)
Respiratory events	SN1	+0.098	+0.059	−0.012	+0.071	+0.991 (1,756)
Respiratory events	SN2	+0.109	+0.060	−0.049	+0.109	+0.501 (2,434)
Respiratory events	SN3	+0.038	+0.034	−0.014	+0.048	+0.947 (11,584)
Respiratory events	SN4	+0.194	+0.237	−0.132	+0.370	+0.827 (3,209)
Respiratory events	SN5	+0.208	+0.255	−0.038	+0.292	+0.964 (8,771)
Limb movements	SN1	+0.149	+0.077	−0.007	+0.083	+0.993 (3,193)
Limb movements	SN2	+0.118	+0.038	−0.003	+0.040	+0.981 (1,176)
Limb movements	SN3	+0.063	+0.015	−0.001	+0.016	+0.982 (1,133)
Limb movements	SN4	+0.055	+0.021	+0.001	+0.020	+0.969 (2,963)
Limb movements	SN5	+0.132	+0.094	−0.016	+0.110	+0.960 (3,085)

Agreement is Cohen's kappa for sleep stages and event F1 for the event tasks. ΔA: mean over the 12 experts of agreement(assisted, pre-fill) minus agreement(manual, pre-fill). Δ_{auto}: the pre-fill's agreement with the assisted minus the manual consensus. Δ_{human}: mean over experts of each expert's manual scoring's agreement with the other 11 experts' assisted minus manual consensus. DiD: Δ_{auto} minus Δ_{human}. Side-taking difference: among the n units (epochs for stages, 0.1-second bins for events) where the two consensuses exist and differ, the share equal to the pre-fill in the assisted minus the manual consensus. These are the values behind Table 1 of the main text; the independent re-implementation reproduced ΔA for all 20 units and Δ_{auto}, Δ_{human} and DiD for sleep stages.

Table S16: Anchoring by expert: each expert's mean ΔA over the five recordings of each task (M1).

Expert	Sleep stages	EEG arousals	Respiratory events	Limb movements
Expert 1	−0.001	+0.043	+0.015	+0.096
Expert 2	+0.166	+0.058	+0.136	+0.128
Expert 3	+0.026	+0.051	+0.134	+0.030
Expert 4	+0.053	+0.081	+0.229	+0.149
Expert 5	+0.038	−0.030	+0.117	+0.076
Expert 6	+0.011	−0.003	+0.070	+0.086
Expert 7	+0.213	+0.014	+0.110	+0.114
Expert 8	+0.021	+0.120	−0.014	+0.124
Expert 9	+0.195	+0.183	+0.156	+0.077
Expert 10	+0.120	+0.090	+0.283	+0.164
Expert 11	+0.060	+0.078	+0.162	+0.095
Expert 12	+0.068	+0.105	+0.153	+0.096

ΔA: assisted minus manual agreement of the expert with the pre-fill (kappa for stages, event F1 for the event tasks). Expert numbers are the dataset's scorer indices; whether the same index denotes the same person in every task is not documented, which the between-task correlations in the text assume.

Table S17: Blind second coding of the 22 FDA summaries with performance data (amendment 005, D1): agreement of each coder with the AI's checked coding, overall and by the coder's exposure.

Field	Coder	All 22 agreement (kappa)	Blind summaries agreement (kappa)	Previously checked agreement (kappa)
Reference	P.T.D.	21/22 (0.937)	11/12 (0.892)	10/10 (1.000)
Reference	D.H.A.N.	18/22 (0.747)	8/10 (0.672)	10/12 (0.780)
Blinding	P.T.D.	22/22 (1.000)	12/12 (1.000)	10/10 (1.000)
Blinding	D.H.A.N.	20/22 (0.853)	8/10 (0.643)	12/12 (1.000)
Pre-scoring	P.T.D.	22/22 (1.000)	12/12 (1.000)	10/10 (1.000)
Pre-scoring	D.H.A.N.	20/22 (0.636)	8/10 (0.000)	12/12 (1.000)

Unweighted Cohen's kappa over the coding sheet's categories (6, 5 and 5). P.T.D. had checked the first 10 of the 22 summaries in the first round and D.H.A.N. the other 12, and each had then seen the AI's codes for them, so each was blind for the other's summaries only. Kappa is unstable where the first coding is nearly constant: pre-scoring was "not said" in 20 of 22 summaries, and on D.H.A.N.'s blind summaries every first code was "not said", so any agreement short of 10 of 10 gives kappa 0.

Table S18: The 8 fields on which a coder's code differed from the first coding, in 5 summaries, and their adjudication from the summary text (amendment 005, D1).

510(k)	Device	Field	First coding	Coder's code (coder, exposure)	Adjudicated
K120450	Sleep Profiler	Reference	Consensus, independence not stated	Panel, daily practice (D.H.A.N. blind)	Consensus, independence not stated
K120450	Sleep Profiler	Blinding	Not stated	Implied by design (D.H.A.N. blind)	Not stated
K142825	Zmachine	Pre-scoring	Not said	Says they did (D.H.A.N. blind)	Not said
K153412	Sleep Profiler	Reference	Consensus, independence not stated	Another algorithm's output (D.H.A.N. blind)	Consensus, independence not stated
K153412	Sleep Profiler	Blinding	Not stated	Implied by design (D.H.A.N. blind)	Not stated
K153412	Sleep Profiler	Pre-scoring	Not said	Not applicable (D.H.A.N. blind)	Not said
K221179	SomnoMetry	Reference	Routine clinical scoring	Not described (P.T.D. blind; D.H.A.N. seen)	Not described
K223539	Dreem 3S	Reference	Consensus, independence not stated	Independent experts, majority (D.H.A.N. seen)	Consensus, independence not stated

"Blind": the coder had not checked that summary in the first round. The adjudication, by the corresponding author, who had not coded, changed 1 code (K221179), to the code both coders had chosen; Table S9 and the Results follow the adjudicated coding. The passages behind every item and both coders' quotations are in the study repository.

Supplementary Figure

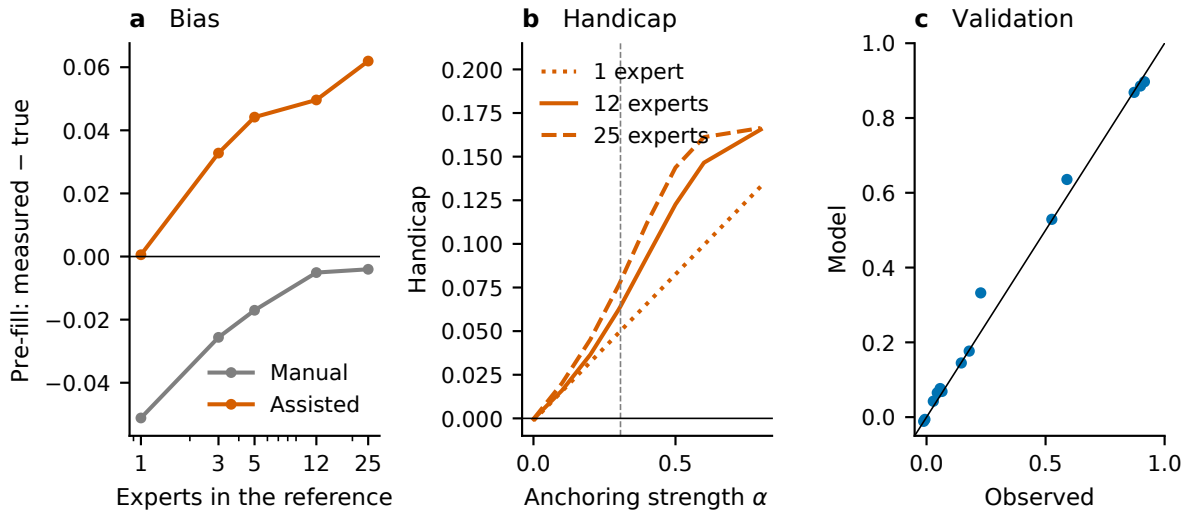


Figure S1: Simulation with the truth known (calibrated model; outputs are illustrations under the model's conditions). (a) The pre-fill's measured minus true accuracy against references of 1 to 25 experts (majority consensus), for manual and assisted references. (b) The handicap in favor of the pre-fill over an equally accurate competitor, against anchoring strength, for references of 1, 12 and 25 assisted experts; the dashed line marks the calibrated α. (c) Model against observed values for the validation targets.

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